

CS339 Autumn 25 Quiz 1 Solution

Q1.
1: True (0.5)
2: False (0.5)

Q2.
A. (car (cdr (list 1 (list 2 3) 4)))
(car (cdr '(1 (2 3) 4)))
(car '((2 3) 4))
'(2 3)
-0.25 for each skipped step
1 mark if fully correct
quote signs are important (-0.25)

B. ((lambda (f) (f 3)) (lambda (x) (* x x)))
((lambda (x) (* x x) 3) 3)
(* 3 3)
9
1 mark if fully correct

Q3. (define (map-reduce f g init lst)
 (if (null? lst)
 init
 (g (f (car lst))
 (map-reduce f g init (cdr lst)))))

2 marks for a correct definition.
-0.5 for skipping each important thing (such as base case)
(argument names are important as given in the Q)

Evaluation:

$$(\text{map-reduce } (\lambda x. (+ x 1)) * 1 '(2 3))$$

$$\Rightarrow (* ((\lambda x. (+ x 1)) 2) \\ (\text{map-reduce } f \ g \ 1 \ '(3)))$$

$$\Rightarrow (* 3 \\ (* ((\lambda x. (+ x 1)) 3) \\ (\text{map-reduce } f \ g \ 1 \ '())))$$

$$\Rightarrow (* 3 \\ (* 4 \\ 1))$$

$$\Rightarrow (* 3 \ 4)$$

$$\Rightarrow 12$$

1 mark for a correct reduction
it's okay to skip one step
> 1 steps skipped : -0.5
wrong

Q4.

A. $1 = \text{SUCC } 0$

$$= (\lambda n. \lambda f. \lambda x. f (n \ f \ x)) (\lambda f. \lambda x. x)$$

$$=_{\beta} \lambda f. \lambda x. f ((\lambda f. \lambda x. x) \ f \ x)$$

$$=_{\beta} \lambda f. \lambda x. f ((\lambda x. x) \ x)$$

$$=_{\beta} \lambda f. \lambda x. \underline{f \ x}$$

$$2 = \text{SUCC } 1$$

$$= (\lambda n. \lambda f. \lambda x. f (n \ f \ x)) (\lambda f. \lambda x. f \ x)$$

$$=_{\beta} \lambda f. \lambda x. f ((\lambda f. \lambda x. f \ x) \ f \ x)$$

$$=_{\beta} \lambda f. \lambda x. \underline{f (f \ x)}$$

1 mark for
each correct
derivation

$$B. (\lambda m. \lambda n. m (\text{SUCC } n)) \ 1 \ 1$$

$$=_{\beta} 1 \ (\text{SUCC } 1)$$

// okay to skip one β -step here

$$= (\lambda f. \lambda x. f \ x) \ (\text{SUCC } 1)$$

$$=_{\beta} \lambda x. (\text{SUCC } 1) \ x$$

$$= \lambda x. (\lambda f. \lambda x. f \ (f \ x)) \ x$$

$$=_{\alpha} \lambda x. (\lambda f. \lambda y. f \ (f \ y)) \ x$$

$$=_{\beta} \lambda x. (\lambda y. x \ (x \ y))$$

$$\approx \lambda x. \lambda y. x \ (x \ y)$$

2 marks for correct reduction

-1 if α -conversion
not applied